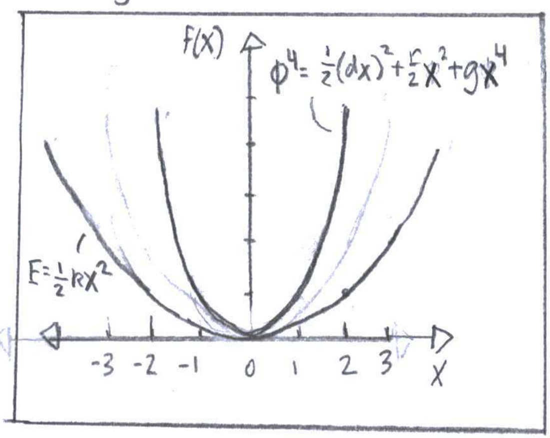
Chapter 4: Perturbation Theory:

Exercise #1:
(pg 175)



"ϕ^4- theory - a prototypical Harmonic Oscillator"

(pg 175) "Action Expansion"

$$S[\phi] = \sum_{k} \left(\phi_k (c_1 + c_2 k \cdot k) \phi_{-k} + c_3 \phi_k h_{-k} \right) + \frac{c_4}{N} \sum_{k_1, \dots, k_4} \phi_{k_1} \phi_{k_2} \phi_{k_3} \phi_{k_4} \delta_{k_1+k_2+k_3+k_4} + O(k^4)$$

where $\phi_i = \frac{1}{\sqrt{N}} \sum_{k} e^{-ik \cdot r_i} \phi(k)$

$$K_{ij} = \frac{1}{N} \sum_{k} e^{-ik(r_i - r_j)} K(k)$$

(pg 175) "ϕ^4- theorem"

$$S[\phi] = -\sum_{ij} \phi_i K_{ij} \phi_j + \sum_{i} \phi_i h_i + \sum_{i} \ln\cosh\left(2 \sum_j K_{ij} \phi_j\right)$$

Term-by-term comparison:

① $\phi_k (c_1 + c_2 k \cdot k) \phi_{-k} = \phi_i K_{ij} \phi_j$

② $c_3 \phi_i h_i = \phi_i h_i$

③ $\frac{c_4}{N} \circ \phi_{k_1} \phi_{k_2} \phi_{k_3} \phi_{k_4} \delta_{k_1+k_2+k_3+k_4} = \ln\cosh\left(2 \sum_j K_{ij} \phi_j\right)$

(From problem) "Expansion"

$$K_{ij} = K(0)\phi(k) + \frac{1}{2} k^2 K''(0) \cdot \phi(k)$$

① $C_1 = K(0)\,\Phi(k)$

② $C_2 = \frac{1}{2} K^2 K''(0)\,\Phi(k)$

③ $C_3 = -1$

④ $C_4 = 4K^2(0)\,\Phi^2(k) - \frac{4}{3} K(0)\,\Phi^4(k)$

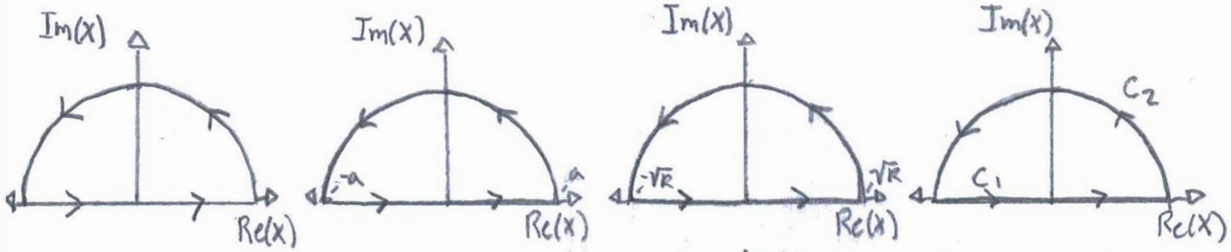
Log (cosh) Series:

$$\ln(\cosh(x)) = \frac{x^2}{2} - \frac{x^4}{12}$$

Exercise #2:
(pg 177)

The correlation function is an expectation to amplitude for fields.

"Complex Contour"



$$\oint F(x)\,dx = \int_{-a}^{a} F(x)\,dx = \int_{-\infty}^{\infty} \frac{e^{ikx}}{k^2+1}\,dx = \oint_{C_1} F(x)\,dx + \oint_{C_2} F(x)\,dx = 2\pi i \sum \operatorname{Res}(f, x)$$

"Residue Theorem"

Cite: Cauchy, A.

"Mémoire sur les intégrales definies, prisis entre des limites imaginaires"

Paris, France (1825)

Exercise #3:
(pg 178)

(4.12) "Green function-inverse Transform"

$$G_0(x) = \frac{1}{L^d} \sum_p e^{-ipx} G_{0,p}$$

$$\cong \int \frac{d^d p}{(2\pi)^d} \frac{e^{-ipx}}{p^2 + r}$$

(4.17) "For dimensions $d > 1$, the integral is divergent"

$$G_0(x) = \int \frac{d^d k}{(2\pi)^d} \frac{1}{p^2 + r}$$

Note: The exercise encounters a critical dimension ($d > 1$).

<u>Two-dimensions ($d=2$):</u>

① <u>Complex integral:</u>

$$G_0(x) \overset{d=2}{=} \int_{-\infty}^{\infty} \frac{d^d k}{(2\pi)^2} \frac{e^{-ikx}}{k^2 + r}$$

② <u>Vertical asymptotes:</u>

$$= \int_{-\infty}^{\infty} \frac{d^2 k}{(2\pi)^2} \frac{e^{-ikx}}{(k - ir^{1/2})(k + ir^{1/2})}$$

asymptotes at $k = \pm\sqrt{+r}$

(3) Residue to Complex Contour:

$$= \frac{1}{(2\pi)^2}\int_{-\infty}^{\infty} dk \left[2\pi i \operatorname*{Res}_{k=\sqrt{-r}} \left[\int dk \, \frac{e^{-ikx}}{k^2 - ir^{1/2}}\right] + 2\pi i \operatorname*{Res}_{k=-\sqrt{-r}}\left[\int dk \, \frac{e^{-ikx}}{k^2+ir^{1/2}}\right]\right]$$

(4) Evaluate one root:

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} dk \, i \cdot \left[\frac{e^{-\sqrt{r}x}}{2i\sqrt{r}}\right]$$

$$= \frac{1}{4\pi}\int_{-\infty}^{\infty} dk \, \frac{e^{-\sqrt{r}x}}{\sqrt{r}}$$

$$\cong \begin{cases} -\frac{1}{2\pi} \ln \frac{\sqrt{r}x}{2} & x \ll 1/\sqrt{r} \\ \frac{1}{2}\left(2\pi\sqrt{r}|x|\right)^{-1/2} e^{-\sqrt{r}x} & x \gg 1/\sqrt{r} \end{cases}$$

Three-dimensions (d=3):

$$G_0(x) \overset{d=3}{=} \int \frac{d^3k}{(2\pi)^3} \frac{e^{-ikx}}{k^2+r}$$

$$= \frac{1}{8\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} d^2k \, \frac{e^{-\sqrt{r}x}}{\sqrt{r}}$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{d^2k}{4\pi} \frac{e^{-\sqrt{r}x}}{\sqrt{r}}$$ Book form, but divergent.

Exercise #4:
(pg 180)

(3.21) "Product over infinite sets"

$$\langle v(x_1)\,v(x_2)\cdots v(x_{2n})\rangle = \sum_{\substack{\text{pairs}\\ \{x_1,\dots,x_{2n}\}}} A^{-1}(x_{k_1}, x_{k_2})\cdots A^{-1}(x_{k,2n-1}\; x_{2n})$$

$$\langle \phi(x)\circ\phi(y)^4\circ\phi(y')^4\circ\phi(x')\rangle_0 = 2^{10-1}\text{ total} \times (10-1)\text{ permutations}$$

$$= 945 \text{ terms}$$

$\boxed{\text{Note: Incorrect typeset in line (4.18)}}$

$$\langle\phi(x)\circ\phi(y)^4\circ\phi(y')^4\circ\phi(x')\rangle = 9G_0(x-x')G_0(0)^4$$

$$+72G_0(x-x')G_0(y-y')G_0(0)^2 + 24G_0(x-x')G_0(y-y')^4$$

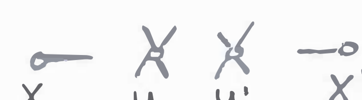
$$+72G_0(x-y)G_0(x'-y')G_0(0)^3$$

$$+288\,G_0(x-y)G_0(x'-y)G(y-y')$$

$$+192\,G_0(x-y)G(y-y')G(x'-y')$$

$$+288\,G(x-y)G(y-y')G(x'-y')$$

Exercise #5:
(pg 181)

The problem uses 

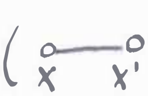
A connection () describes a 2-point correlation. with an integral.

Diagram #1: $9\,G(x-x')\,G_0(0)^4$

Diagram #2: $72\;G(x-x')\cdot G(y-y')\cdot G_0(0)^2$

Diagram #3: $24\,G_0(x-x')\,G_0(y-y')^4$

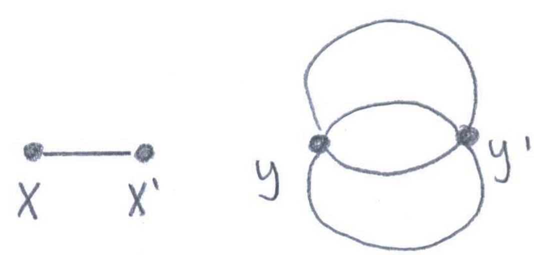


Diagram #4: $72\;G_0(x-y)\,G_0(x'-y)\,G_0(0)^3$

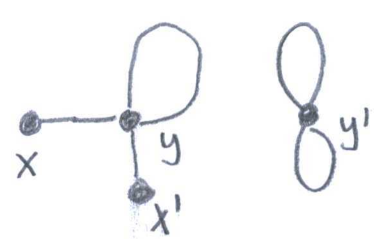


Diagram #5: $288\;G(x-y)\,G_0(y-y')\,G_0(x'-y)$

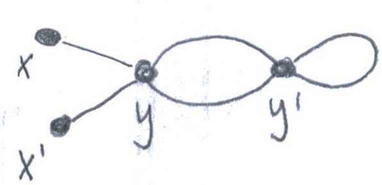


Diagram #6: $192\, G_0(x-y)G_0(y-y')G_0(x'-y)$

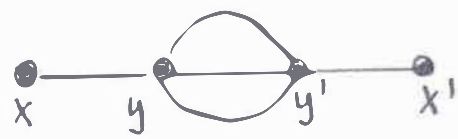
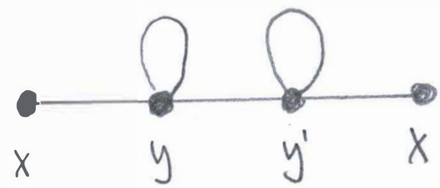


Diagram #7: $288\, G(x-y)G(y-y')G(x'-y')$



$$\boxed{\text{The correlation function is on line (4.15).}}$$

Exercise #6: (pg 134)

(4.14) "Functional expectation value"

$$\langle X[\phi]\rangle \simeq \frac{\sum_{n=0}^{\infty}\frac{(-g)^n}{n!}\left\langle X[\phi]\left(\int d^dx\,\phi^4\right)^n\right\rangle_0}{\sum_{n=0}^{\infty}\frac{(-g)^n}{n!}\left\langle\left(\int d^dx\,\phi^4\right)^n\right\rangle_0}$$

$$\simeq \sum_{n=0}^{n_{max}} X^{(n)}$$

$$\langle X[\phi]\rangle \simeq \frac{\sum_{n=0}^{2}\frac{(-g)^n}{n!}\left\langle X[\phi]\left(\int d^dx\,\phi^4\right)^n\right\rangle_0}{\sum_{n=0}^{2}\frac{(-g)^n}{n!}\left\langle\left(\int d^dx\,\phi^4\right)^n\right\rangle_0}$$

$$= \frac{N_0 + N_1 + \frac{1}{2}N_2}{1 + D_1 + \frac{1}{2}D_2}$$

Binomial Expansion:

$$\boxed{(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2}$$

$$\cong N_0 - N_1 - N_0 D_1 + \frac{1}{2} N_2 - N_1 D_1 + N_0 D_1^2 - \frac{1}{2} N_0 D_2$$

$$N_0 = G_0(x-x')$$

$$N_1 = 3G_0(x-x')G_0^2(0) + 12\,G_0(x-y)G_0(0)G(y-x)$$

$$N_2 = 9G_0(x-x')G_0(0)^4 + 72\,G_0(x-x')G_0(y-y')G_0(0)^2$$

$$+ 24\,G_0(x-x')G_0(y-y')^4$$

$$+ 72\,G_0(x-y)G_0(x'-y)G_0(0)^3$$

$$+ 288\,G_0(x-y)G_0(x'-y)G(y-y')$$

$$+ 192\,G_0(x-y)G_0(y-y')G(x'-y)$$

$$+ 288\,G(x-y)G(y-y')G(x'-y)$$

$$D_1 = 3G_0^2(0)$$

$$D_2 = 9G_0^4(0) + 72G_0^2(y-y') + 24G_0^4(y-y')$$

$$G^{(0)} = N_0$$

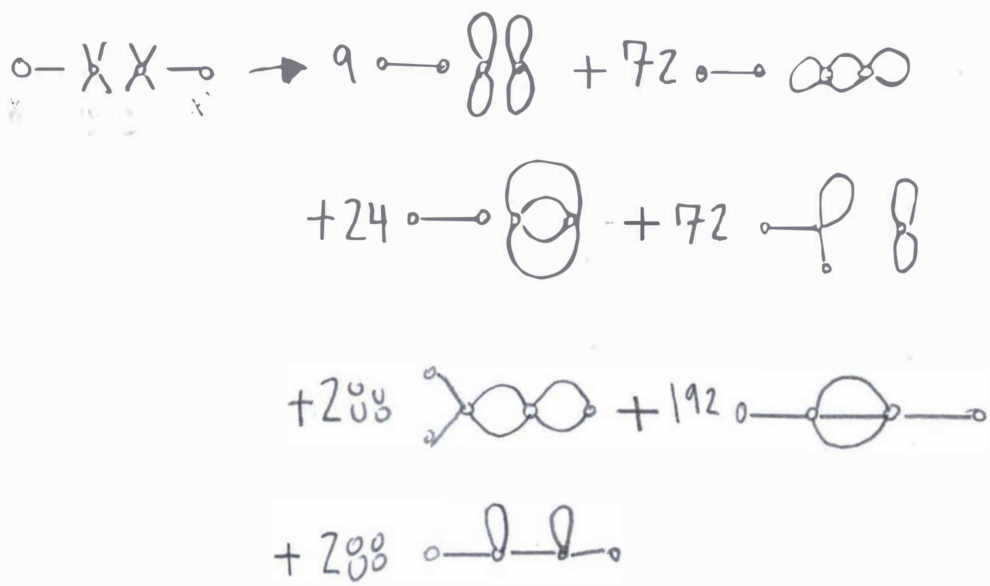
$$G^{(1)} = N_1 - N_0 D_1$$

$$G^{(2)} = \frac{1}{2} N_2 - N_1 D_1 + N_0 D_1^2 - \frac{1}{2} N_0 D_2$$

Exercise #7

(pg 136)

Figure 4.2 : Greens Function



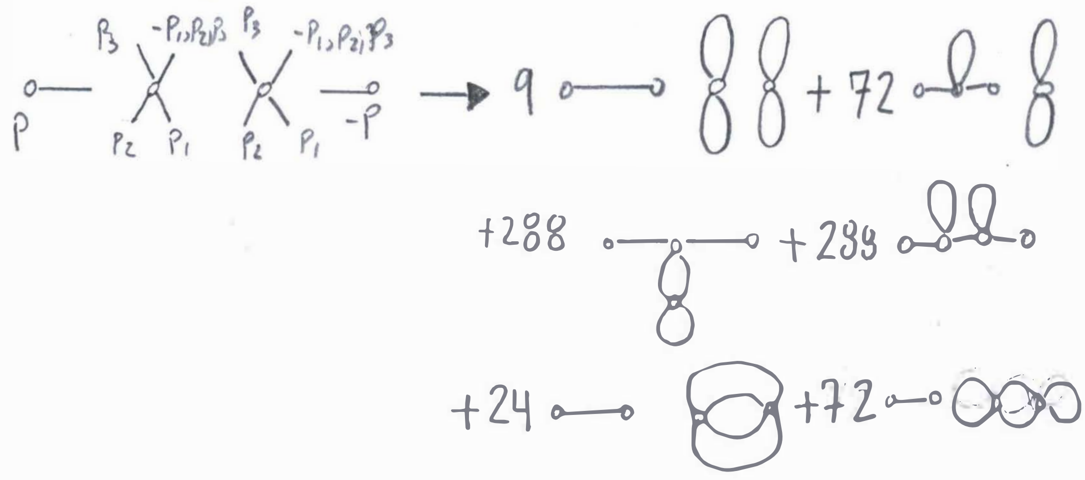
(4.12) "Greens function - modified"

$$G_0(x-x') \approx \int \frac{d^dp}{(2\pi)^d} \frac{e^{-ip(x-x')}}{p^2+r}$$

People drew pictures for integrals

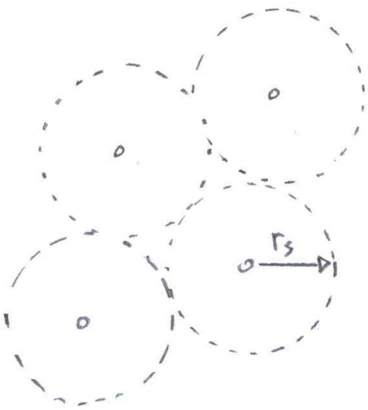
Others drew pictures for momentum in the same function.

p-diagrams:



$+192$ 

Exercise #8:
(pg 192)

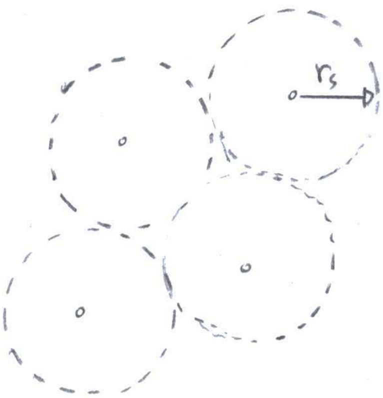


"Ball packing"

Cite: Keplar. J

"Strena seu de nive sexangula"

Francofvrti, Germany (1611)



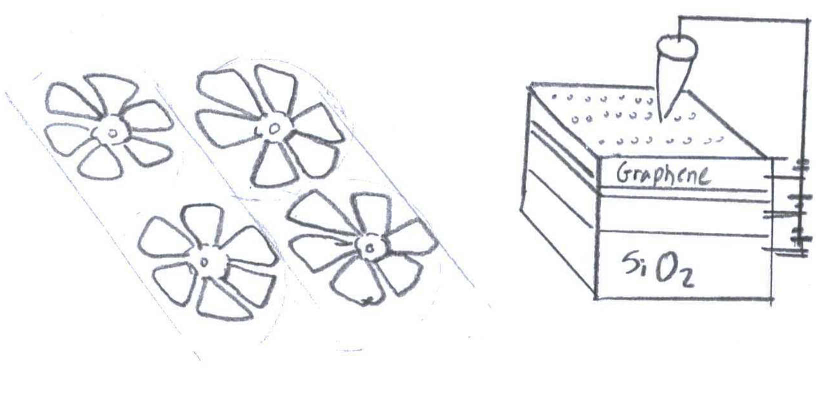
$$r_s = \frac{\sqrt[3]{\text{Average Distance}}}{\text{Bohr radius}}$$

"Fermion gas"

Cite: Cornell, E., Wieman E.

"Observation of Bose Einstein condensation in a dilute atomic vapor"

Colorado (1995)



"Wigner Crystal"

Cite: Grimes, C. Adams G.

"Evidence for a liquid-to-crystal phase transition in a classical, two dimensional Sheet"

Bell Labs, (1979)

Second-order expansion:

(pg 139) "Free Energy"

$$F = -T \ln Z$$

$$= -T \ln \left(\int D\psi \, e^{-S[\phi]} \right)$$

$$= -T \ln \left(\int D\psi \, e^{-S_0[\psi] - S_{int}[\psi]} \right)$$

$$= -T \ln \left(\int D\psi \, e^{-S_0[\psi]} \left(1 - S_{int}[\psi] + \frac{1}{2} S_{int}[\psi]^2 \right) \right)$$

$$= -T \ln \left(\int D\psi \, e^{-S[\psi]} - \int D\psi \, e^{-S_0[\psi]} \cdot S_{int}[\psi] - \right.$$

$$\left. - \int D\psi \, e^{-S_0[\psi]} \cdot \frac{1}{2} S_{int}[\psi]^2 \right)$$

$$= -T \ln \left(Z_0 \left(1 - \langle S_{int}[\psi] \rangle + \frac{1}{2} \langle S_{int}^2 \rangle \right) \right)$$

$$\boxed{\begin{array}{l} \underline{\text{Logarithm Expansion}} \\ \ln(1+x) = x - \left(\frac{1}{2}\right) x^2 + \cdots \end{array}}$$

$$= -T \left(\underbrace{\ln Z_0}_{F^{(0)}} \underbrace{- \langle S_{int} \rangle_0}_{F^{(1)}} + \underbrace{\frac{1}{2} \langle S_{int}^2 \rangle_0 - \frac{1}{2} \langle S_{int} \rangle_0^2}_{F^{(2)}} + \cdots \right)$$

$$F^{(2)} = -\frac{T}{2} \left(\langle S_{int}^2 \rangle_0 - \langle S_{int} \rangle^2 \right)_0$$

Disconnected Diagrams:

(4.19) "Zeroth order perturbation"

$$F^{(0)} = -T \sum_{p\sigma}' \ln\left(1 + e^{-\beta\left(\frac{p^2}{2m} - \mu\right)}\right)$$

$$\cong -\frac{2}{5} N\mu$$

(4.21) "First order perturbation"

$$F^{(1)} = \frac{T^4}{2L^3} \left\langle \sum_{pp'q,\sigma\sigma'} \bar{\psi}_{p+q\sigma} \bar{\psi}_{p'-q\sigma'} V(q) \psi_{p'\sigma'} \psi_{p\sigma} \right\rangle_0$$

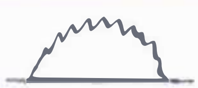
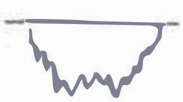
$$= -\frac{T^2}{L^3} \sum G_p V(p-p') G_{p'}$$

(pg 152) "Second order perturbation"

$$F^{(2)} = -\frac{T}{2}\left(\frac{T^3}{2L^3}\right)^2 \left\langle \left(\sum_{pp'q} \bar{\psi}_{p+q\sigma} \bar{\psi}_{p'-q\sigma'} V(q) \psi_{p'\sigma'} \psi_{p\sigma} \right)^2 \right\rangle_0$$

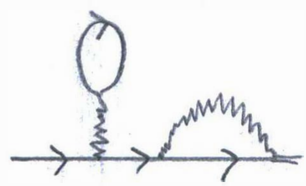
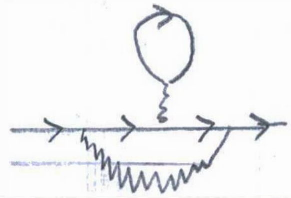
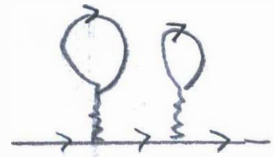
$F^{(2)}$ is the only relatable term to $V(q=0)$ for "a wavy line" on page 190. The line indicates a zero evaluation by $V(q=0) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = 0$

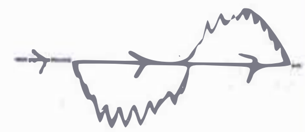
Exercise #9
(pg 202)

Shape	Purpose
	Green Function = $\frac{1}{\lvert r \rvert}$
	Coefficient (1/N)
	Green function difference = $\frac{1}{\lvert r-r' \rvert}$
	positive (spin up)
	negative (spin down)

"Non-crossing approximation"

Papers challenged $O(N^2)$ degeneracy in a matrix for the computer by N.C.A.

Diagrams:	Functions
	$\frac{g}{L^d}\left(\frac{1}{N}\sum G_{0,p,p'} \circ G_{0,p} \circ G_{0p'} \circ G_{0,p+p'}\right)$
	$\frac{g}{L^d}\left(\frac{1}{N}\sum G_{0p} \circ G_{0,p-p'} \circ G_{0,p+p'} \circ G_{0p'}\right)$
	$\frac{g}{L^d}\left(\frac{1}{N}\sum_i G_{0p}, G_{0,p'p} \circ G_{0,p-p'} \circ G_{0,p'}\right)$

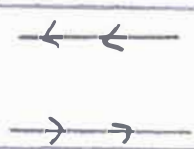
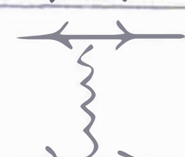
	$\frac{g}{L^d}\sum_p G_{p,0} \circ G_{0,-p-p'} \circ G_{0,p+p'} \circ G_{0,p'}$
	$\frac{g}{L^d}\sum_p G_{0,p} \circ G_{0,p+p'} \circ G_{0,p'+p} \circ G_{0,p'} \circ G_{0,p''}$
	$\frac{g}{L^d}\sum G_{0,p} G_{0,p+p''} \circ G_{0,p'} \circ G_{0,p'+p''} G_{0,p''}$

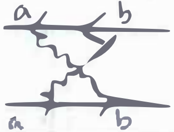
(4.44) "Non-crossing approximation"

$$\Sigma_p \overset{NCA}{=} -\frac{g}{L^d}\sum G_{p'}$$

∴ Unsure on the truth about efficiency against $O(N^2)$ computational operations. I bet for the d-dimensional matrix.

Exercise # 10:
(pg 205)

Schematic	Function
	"Greens Function" = $G(p)$
	Two Greens functions $G(p)$ and $G(p')$
	Functions multiplied $G(p) \circ G(p')$

	$G(p,a)\circ G(p',b) + G(p,b)\circ G(p',a)$
	$G(p,a)\circ G(p',b) + G(p,b)\circ G(p',a)$ $+ G(p,b)\, G(p',b)$

Problem 4.6.1: Technical aspects of diagrammatic perturbation theory:

a) $\langle \phi(x)\,\phi^4(y)\,\phi^4(y')\,\phi(x')\rangle$ = Exercise #6 - pg 134

b) Exercise #6 - pg 134

c) Exercise #7 - pg 136

Problem 4.6.2: Self-consistent T-matrix approximation:

a) (4.69) "T-matrix"

$$T_n = \langle H_{imp} + H_{imp}\, G_{0,n}(0)\, H_{imp} + H_{imp}\, G_{0,n}(0)\, H_{imp}\, G_{0,n}(0)\, H_{imp} + \cdots \rangle$$

where $H_{imp} = \gamma \sum_i \delta(r - R_i)$

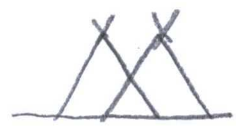
(4.22) "Green function of the electron gas"

$$G_p = \frac{1}{i\omega - \frac{p^2}{2m} + \mu} = \frac{1}{i\omega - H_0 - H'_{imp}}$$

$$T_n = \langle \gamma \sum_i \delta(r-R_i)\left(1 + \gamma \sum_i (r-R_i)\, G_{0,n}(0)\right)\rangle$$

$$G_n = G_{0,n} + G_{0,n} \langle \gamma (1 + \Sigma_i G_{0,n}) \rangle G_{0,m}$$

b) <u>Crossing impurity lines:</u>

 the author showed a diagram for difference in a Greens function. In a one-to-one system, a line is an integral. Where two integrals cancel and "leave out" an impurity.

c) $T^{\pm}(\epsilon) \equiv \hat{T}(i\omega_n \to \epsilon \pm i\delta)$

$$T^{+}(\epsilon) - T^{-}(\epsilon) = \hat{T}(i\omega_n \to \epsilon + i\delta) - \hat{T}(i\omega_n \to \epsilon - i\delta)$$

$$= \frac{1}{\gamma^{-1} + G_n^{+}(0)} - \frac{1}{\gamma^{-1} - G_n(0)}$$

$$= \frac{G^{+}(\epsilon, 0) - G^{-}(\epsilon, 0)}{(\gamma^{-1} + G_n^{+}(0))(\gamma^{-1} - G_n(0))}$$

$$= T^{+}\left(G^{+}(\epsilon,0) - G^{-}(\epsilon,0)\right) T^{-}$$

$$= |T^{+}(t)|^{2} \nu$$

Problem 4.6.3 - Kondo Effect - Perturbation :

a) (2.50) "sd-Hamiltonian"

$$\hat{H}_{sd} = \sum_{k\sigma} \epsilon_k \circ c^{\dagger}_{k\sigma} c_{k\sigma} + 2J\hat{S}_d \circ \hat{S}(0)$$

where $\hat{S}(0) = \sum_{kk'\sigma\sigma'} c^{\dagger}_{k\sigma} \sigma_{\sigma\sigma'} c_{k'\sigma'}$

(from problem) "Hamiltonian"

$$\hat{H} = \sum_{k\sigma} \epsilon_k c^{\dagger}_{k\sigma} c_{k\sigma}$$

$$H_{imp} = 2J\hat{S} \circ S(r=0)$$

(from problem)

$$\frac{1}{2\tau} = \pi \frac{n_{imp}}{L^d} \sum \langle |\langle k,\sigma| \hat{H}_{imp} |k',\sigma'\rangle|^2 \rangle$$

$$= \pi\, c_{imp} \circ J^2 L^{-d} \sum_{\sigma'} \langle\langle \sigma| S \circ \sigma |\sigma'\rangle^2 \rangle$$

$$= \pi\, c_{imp} \cdot \nu\, J^2 S(S+1)$$

If $L=1$ and $\delta(t, \epsilon_k)=1$

also $\langle\langle \sigma | S\cdot\sigma \, S\cdot\sigma | \sigma \rangle\rangle$

$= S(S+1)$

b) (from problem) "T-matrix"

$$\hat{T}^{(2)} = \hat{H}_{imp}(\epsilon^+ - \hat{H}_0)\hat{H}_{imp}$$

$$Re\langle k', \sigma' | \hat{T}^{(2)} | k, \sigma \rangle$$

$$= Re\langle k', \sigma' | \hat{H}_{imp}(\epsilon - H_0)\hat{H}_{imp} | k, \sigma \rangle$$

$$= Re\langle k', \sigma' | \Big(J\sum (S\cdot\sigma) c_p^\dagger | \Omega \rangle - J\sum (S\cdot\sigma) c_p^\dagger c_k c_p | \Omega \rangle \Big) | k, \sigma \rangle$$

$$= J^2 (S\cdot\sigma)(S\cdot\sigma) \frac{\sum \langle \Omega | c_p^\dagger c_p | \Omega \rangle}{\epsilon_k - \epsilon_p} + J^2 (S\cdot\sigma)(S\cdot\sigma) \frac{\sum \langle \Omega | c_p c_k^\dagger c_p c_p^\dagger c_k c_p | \Omega \rangle}{\epsilon_k - \epsilon_p}$$

$$= J^2 \sum \frac{1}{\epsilon_k - \epsilon_p} (S\cdot\sigma)^2 \left[\Theta(-\xi) - \Theta(\xi) \right]$$

$$= J^2 \sum \frac{1}{\epsilon_k - \epsilon_p} \left[S(S+1) - (2n_F(\xi) - 1) S\sigma_{\sigma\sigma'} \right]$$

when $\Theta(-\xi) = n_F(\xi)$

$\Theta(+\xi) = 1 - n_F$